

Data Analytics Portfolio

# Seasonal Demand Analysis

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Using patterns in sales data to inform inventory planning.

# Context and the question

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## The situation

Management is concerned that inventory planning is being done haphazardly.

## The question

Can we use patterns in sales data to inform inventory planning decisions?

## Scope

Solely focused on patterns in existing product and customer base.

# Approach

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01

## Data

Two years of sales data from 2024 to 2025.

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02

## Analysis

Patterns and trends in monthly sales.

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03

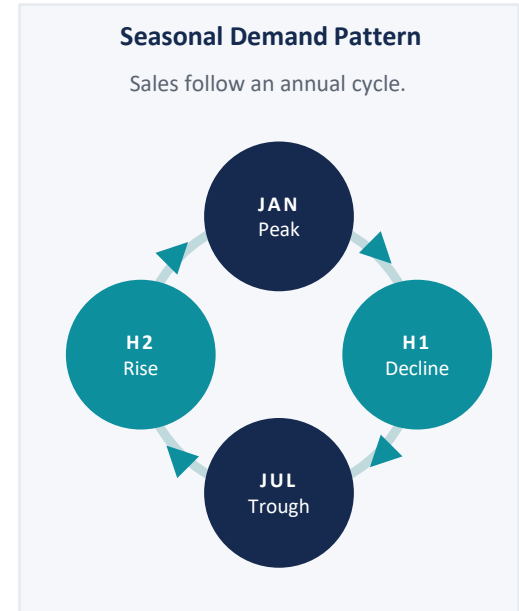
## Segmentation

Demand impact by product category: peripherals, smart watches, and tablets.

# What the data showed

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- Peak in January, trough in July.
- Sales trend down 1<sup>st</sup> half of year, trend up in 2<sup>nd</sup> half.



## What the data showed (cont.)

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### Peripherals Impact

Peripheral products account for the largest proportion of seasonal declines.

| Product Category | Peak to Trough % | Peak to Trough Units | % of Aggregate Decline |
|------------------|------------------|----------------------|------------------------|
| Peripheral       | 52.23%           | 199.5                | 63.23%                 |
| Smart Watch      | 51.03%           | 43.5                 | 13.79%                 |
| Tablet           | 55.71%           | 72.5                 | 22.98%                 |

# Recommendations

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## Seasonal Planning

**Data insight:** Sales follow a seasonal cycle.

**Takeaway:** Inventory planning should account for seasonal variation.

**Action Step:** Develop seasonal inventory targets.

## Forecast Peak Peripherals

**Data insight:** Peripherals make up the largest aggregate peak-to-trough decline.

**Takeaway:** Peripherals may have a stronger impact on seasonal inventory.

**Action Step:** Develop a peak-season demand forecast for peripherals to minimize stockout and carrying costs.

# Limitations

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01

This study only looked at product category. More research is needed to apply findings to specific products.

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02

Findings are based on historical data. Changes in factors that affect sales patterns will require findings to be reassessed with more current data.

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03

This case study uses synthetic data to demonstrate analytical approach. It is not intended to make recommendations for live business practice.